

HOT / COLD / DELTA SEMANTICS AUDIT

From raw PRISM temperatures to the final CONUS map

OVERALL VERDICT

F. MULTIPLE INCONSISTENCIES EXIST

The scientific calculations, stored values, raster sign, and current CONUS pixel colors are correct. The inconsistencies are representational: one incomplete current legend, ten historical reversed palettes, and mixed hot/warm terminology.

Diagnostic report only - no algorithm or production output was changed

Prepared 24 September 2026 | Evidence hashes retained in the audit manifest

COLD +
WARM -

What is correct - and what is not

PASS

Tail assignment: lower TMIN is cold; upper TMAX is warm.

PASS

Delta algebra: every audited value is S cold minus S warm.

PASS

Current CONUS values and pixel colors preserve the sign.

FAIL

Repository-wide visual convention is not uniform.

Core truth table

Condition	Delta	Meaning	Current color
S cold > S warm	> 0	Cold synchrony stronger	Blue
S cold = S warm	= 0	Equal tail synchrony	Near-white
S cold < S warm	< 0	Warm synchrony stronger	Red

Why the verdict is F

1. The standalone CONUS Delta colorbar says only 'Median Spearman synchrony'; its endpoints are not semantically named.
2. Ten historical Delta or bottom-minus-top displays use RdBu_r, assigning positive/cold to red and negative/warm to blue.
3. 'Hot' and 'warm' denote the same upper-TMAX calculation but vary across outputs.

2.22e-16

Largest real-pair recomputation error.

1,846

Relationships independently recomputed for one western pixel.

70.5%

CONUS cells with negative Delta.

980

Offline tests passed after the audit.

Bottom line: the national scientific numbers do not need a sign reversal. The remediation decision concerns visual communication and historical continuity.

The complete production path

Every arrow below was traced to code and checked against retained values



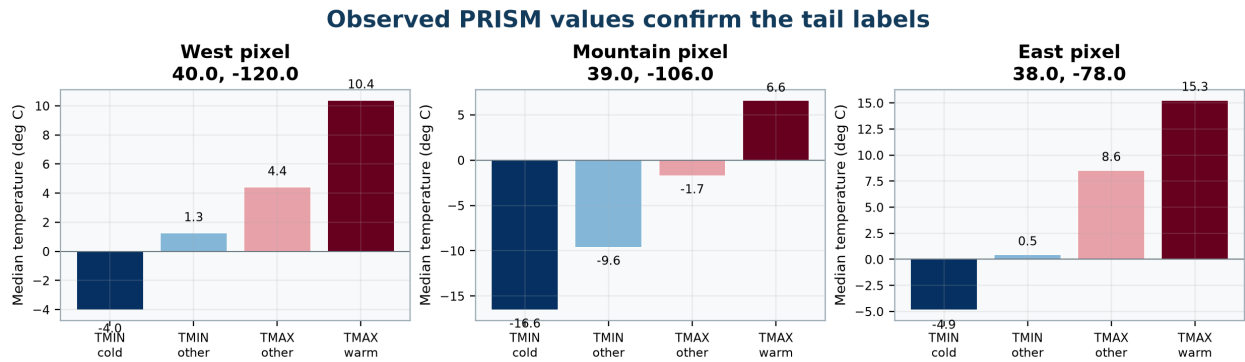
Functions and variables

Stage	Function	Relevant variables
temperature acquisition	load_prism_cube	tmin, tmax
frozen input	acquire / _open_input	cube.tmin, cube.tmax
variable routing	_select_inputs	lower=tmin, upper=tmax
tail state	_precomputed_tail_state	lower <= q; upper > q
pair synchrony	_batched_one_tail_spearman	cold, warm, counts
pair Delta	local_synchrony_pairs	delta_s = cold - warm
pixel collapse	nonstacked_synchrony_summary	delta_pair_median
tiled execution	tiled_nonstacked_synchrony_summary	tile summaries
GeoTIFF output	_write_rasters	delta_pair_median_100km.tif

Audit boundary: the scientific path passes end to end. The identified divergence begins in visual encoding and legend semantics, not in the numerical pipeline.

Cold really is low; warm really is high

Observed PRISM pixels - no synthetic fallback



Literal implementation

Cold: TMIN_A <= median(TMIN_A) AND TMIN_B <= median(TMIN_B)

Warm: TMAX_A > median(TMAX_A) AND TMAX_B > median(TMAX_B)

q: 0.5
Window: 91 daily labels ending 2024-01-30
Ties: average ranks
Missing: paired removal before thresholds
Pair rule: both members must pass their own threshold

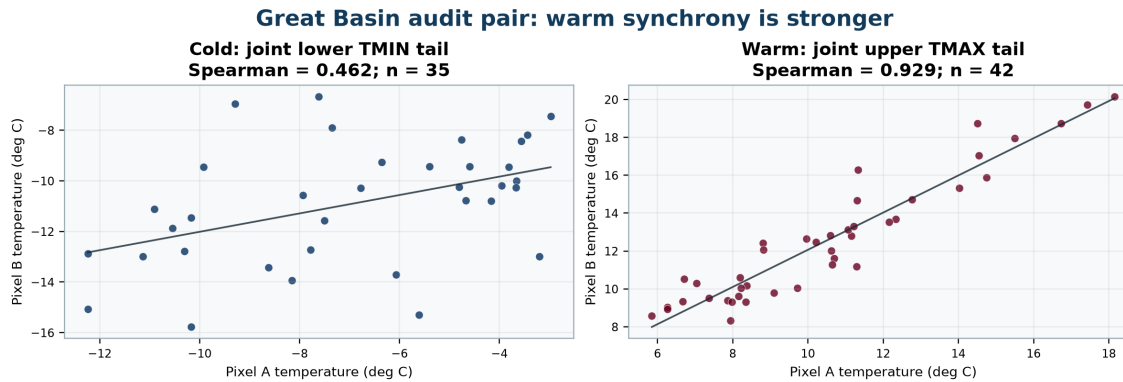
Pixel	Cold median	Warm median	Cold n	Warm n	Pass
West	-4.04 C	10.39 C	46	45	YES
Mountain	-16.55 C	6.63 C	46	45	YES
East	-4.87 C	15.28 C	46	45	YES

PASS

Every observed example places cold dates below and warm dates above its local reference distribution.

Five observed relationships recomputed by hand

Direct SciPy Spearman from the selected dates versus stored production values



Worked Great Basin pair

Pixel A: 39.458, -116.958

Pixel B: 39.000, -116.000

Distance: 97.01 km

S cold = 0.462185

S warm = 0.929017

Delta = -0.466832

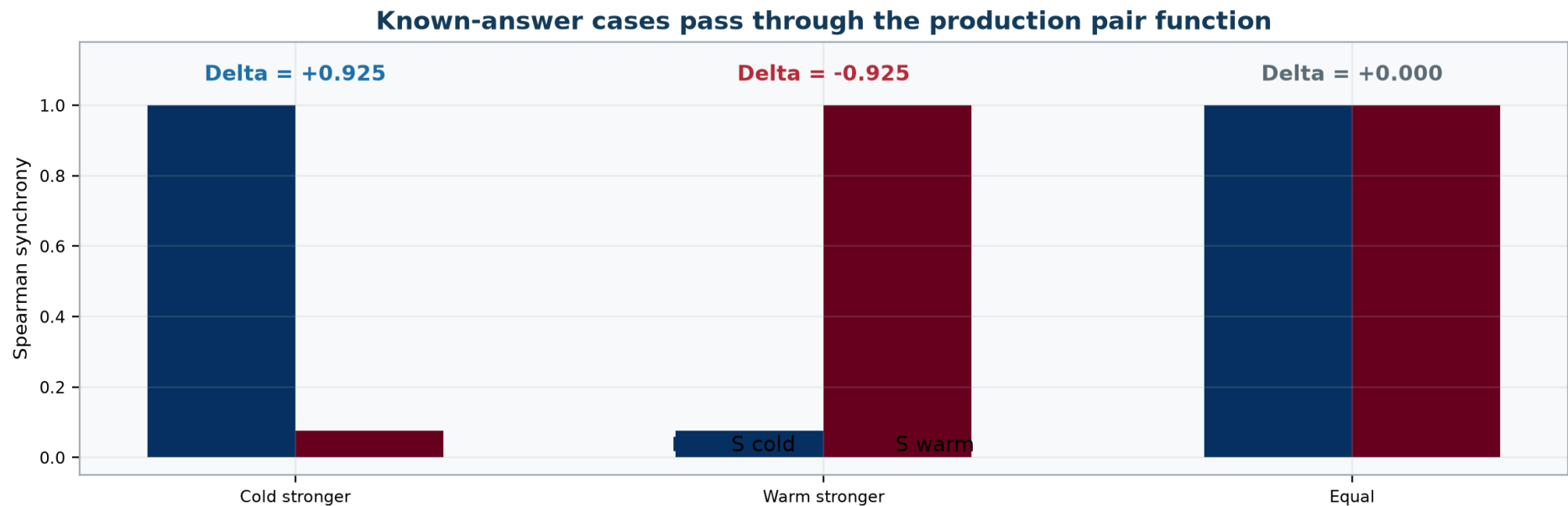
Interpretation: warm synchrony is stronger by 0.467.

Region	km	S cold	S warm	Delta	Max error
Pacific Northwest	99.2	0.725	0.266	+0.459	1.1e-16
Great Basin	97.0	0.462	0.929	-0.467	1.1e-16
Rockies	49.7	0.609	0.952	-0.342	2.2e-16
Plains	67.3	0.958	0.974	-0.016	1.1e-16
East	49.8	0.939	0.956	-0.017	1.1e-16

PASS: all entering dates, temperatures, percentiles, within-tail ranks, and flags are retained in pair_audit_observations.csv. Maximum stored-versus-direct error is 2.22e-16; Delta algebra error is 0.

Three cases with an obvious answer

All cases ran through local_synchrony_pairs - the same production function used for CONUS



A Cold stronger

S cold = 1.000
S warm = 0.075
Delta = +0.925

B Warm stronger

S cold = 0.075
S warm = 1.000
Delta = -0.925

C Equal

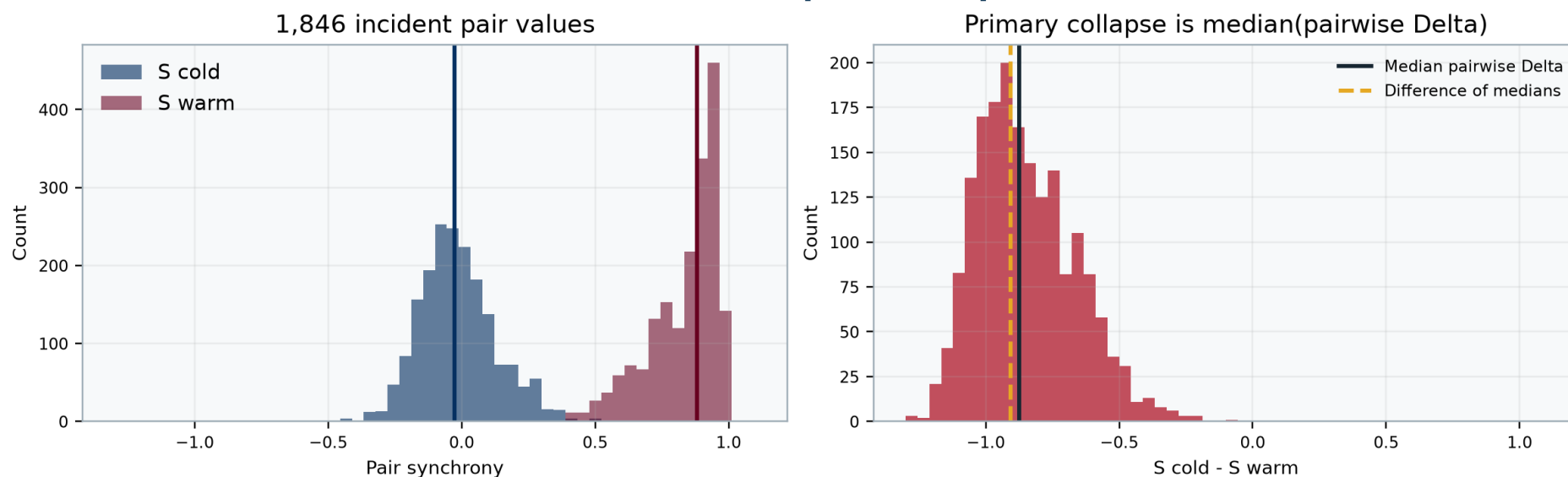
S cold = 1.000
S warm = 1.000
Delta = +0.000

PASS: positive Delta labels cold stronger; negative Delta labels warm stronger; equal synchrony produces exactly zero.

One focal pixel, 1,846 pair relationships

The primary map uses $\text{median}(S_{\text{cold}} - S_{\text{warm}})$, not $\text{median}(S_{\text{cold}}) - \text{median}(S_{\text{warm}})$

Western focal-pixel collapse



-0.025902

Median S cold

0.881413

Median S warm

-0.875348

Primary median pairwise Delta

-0.907315

Separate difference of medians

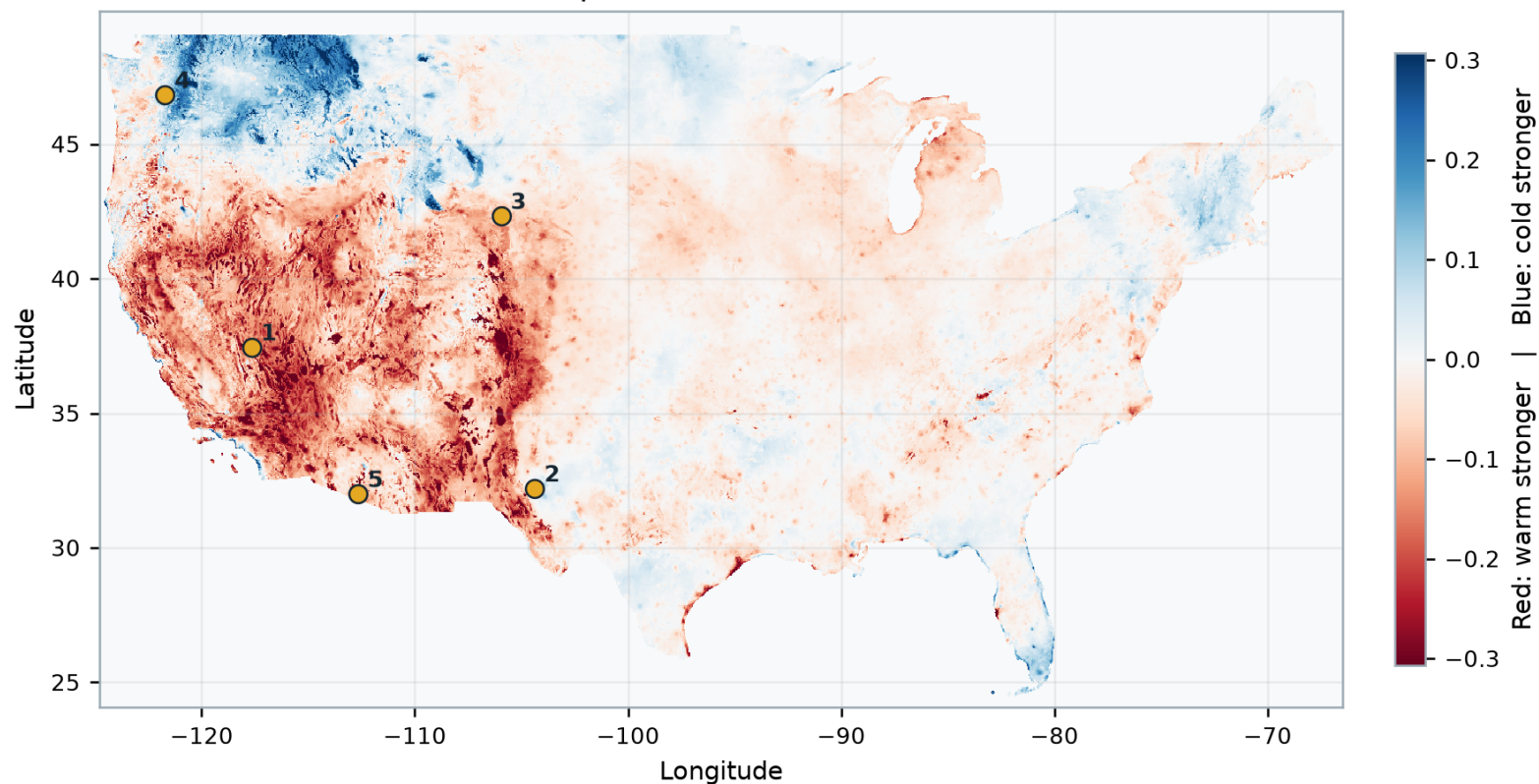
PASS: NetCDF error = 0.0e+00; float32 GeoTIFF error = 2.55e-09. The two collapse formulas differ by 0.031967, so preserving their names is scientifically necessary.

The current CONUS map is colored correctly

Delta = S cold - S warm | red negative/warm stronger | blue positive/cold stronger

Current CONUS values and colors are semantically correct

Median pairwise Delta at 100 km



0.910874

National median S cold

0.928470

National median S warm

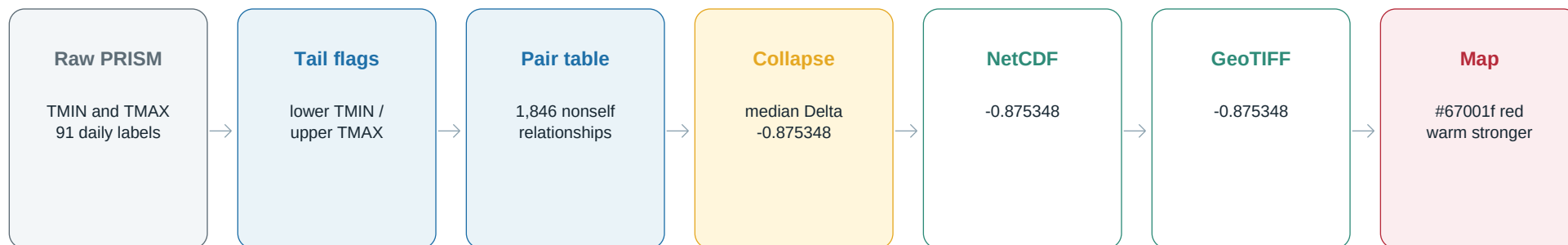
-0.017335

National median pairwise Delta

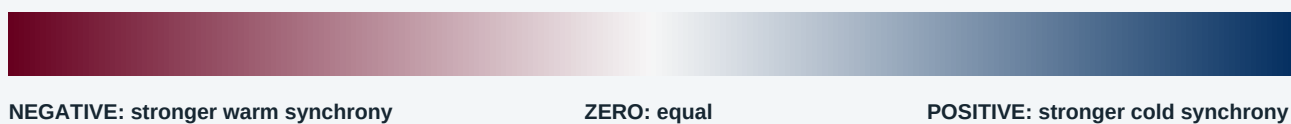
70.5% negative

Algebraically consistent with warm > cold

One western pixel from temperatures to displayed color



Colorbar contract



RED = WARM

for the current CONUS Delta map

Figure-legend finding: the pixels obey this contract, but the standalone colorbar is labeled only 'Median Spearman synchrony.' A reader should not need external prose to infer that red means warm stronger and blue means cold stronger.

Five real spot checks on the prominent red feature

Every sampled red cell is negative and means stronger warm synchrony

#	Latitude	Longitude	Median cold	Median warm	Pairwise Delta	Color	Interpretation
1	37.458	-117.667	-0.026	0.881	-0.875	Red	warm synchrony stronger by about 0.875
2	32.208	-104.375	0.621	0.897	-0.289	Red	warm synchrony stronger by about 0.289
3	42.333	-105.958	0.659	0.893	-0.233	Red	warm synchrony stronger by about 0.233
4	46.833	-121.708	0.116	0.364	-0.237	Red	warm synchrony stronger by about 0.237
5	32.000	-112.667	0.675	0.947	-0.271	Red	warm synchrony stronger by about 0.271

Strongest audited western cell

Location: 37.458, -117.667
Median S cold: -0.025902
Median S warm: 0.881413
Median pairwise Delta: -0.875348

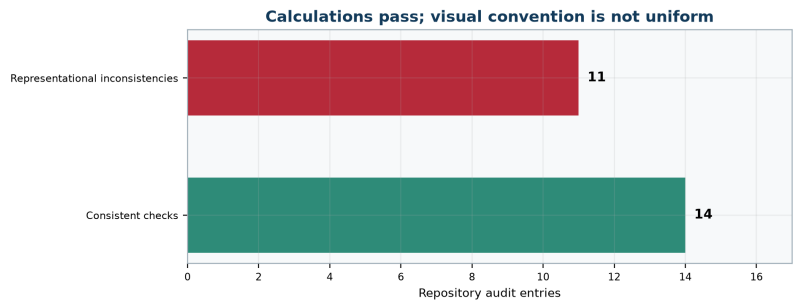
Scientific interpretation

At these cells, warm-tail TMAX synchrony exceeds cold-tail TMIN synchrony. The red western feature does not mean 'hot temperature.' It means stronger spatial synchrony during locally warm-tail days, by the displayed Delta magnitude.

PASS: values, signs, colors, and interpretations agree for every western spot check. Causal or regime claims remain outside the scope of this semantics audit.

The visual convention changed across generations

11 representational inconsistencies; no warm-minus-cold implementation found



Classification

- Algorithmic: PASS
- Variable routing: PASS
- Delta sign: PASS
- Spatial collapse: PASS
- Raster values: PASS
- Current CONUS colors: PASS
- Current standalone legend: FAIL
- Repository-wide palette convention: FAIL
- Terminology uniformity: FAIL

File	Context	Finding
run_nonstacked_conus_baseline.py	CONUS colorbar label	generic label omits red=warm stronger and blue=cold stronger
recipe: climate_tail_dep_center.md	legacy recipe plot	negative blue; positive red
ex: median_split_synchrony_demo.py	legacy synthetic climate Delta viewer	negative blue; positive red
ex: real_prism_median_split_synchrony.py	legacy real-PRISM Delta viewer	negative blue; positive red
ex: climate_synchrony_cube_panel_demo.py	legacy climate Delta panel	negative blue; positive red
ex: center_pixel_synchrony_walkthrough.py	legacy center-pixel walkthrough	negative blue; positive red
recipe: s2_tail_dep_center.md	legacy bottom-minus-top recipe	negative blue; positive red
recipe: s2_tail_dep_manual.md	legacy manual plot	negative blue; positive red
build_synchrony_phase2_atlas.py	historical Phase 2 Delta atlas	negative blue; positive red
build_synchrony_surface_atlas_phase2.py	historical Delta atlas	negative blue; positive red
build_relational_convolution_report.py	historical Delta report	negative blue; positive red

Verdict F reflects multiple communication inconsistencies, not multiple numerical failures. Existing values should not be negated or recomputed as a remedy.

What the team can safely say now

And what should wait for a separately approved remediation

Safe conclusions

- Cold is the joint lower local tail of PRISM TMIN.
- Warm/hot is the joint upper local tail of PRISM TMAX.
- Delta is S cold minus S warm.
- Positive Delta means cold synchrony is stronger.
- Negative Delta means warm synchrony is stronger.
- The current CONUS numbers, rasters, and pixel colors preserve this meaning.
- The western red feature is warm-synchrony stronger.

Proposed remediation - not executed

1. Define one shared Delta palette contract.
2. Add semantic endpoint labels to the CONUS colorbar.
3. Review only synchrony-specific RdBu_r uses.
4. Canonicalize 'warm'; retain 'hot' as a documented alias.
5. Regenerate affected figures and PDFs only after approval.
6. Version historical recolors because human interpretation changes even when values do not.

Verification record

Five real pairs: PASS | Three known-answer cases: PASS | One-pixel NetCDF and GeoTIFF trace: PASS | Five western cells: PASS | 32 focused tests: PASS | Full offline suite: 980 passed, 5 skipped, 419 deselected | Compilation and diff checks: PASS

Evidence identity

Observed PRISM input SHA-256: 7ce255e9e09bebb824a6a7f6326a8ce4de18576148eaa837c89f2852c071092f
CONUS result SHA-256: f3fbced095c324ba752a8d4a25c0b67da4e2935b44f534dcf30385b4f96d5709
Audit manifest SHA-256: aa77b41105de70566711b51432ac9155718e5df3ba6ed5d6617d85ecab45b04d
Generated report SHA-256 is recorded after build in clean-report-assets/report_manifest.json.